LESSON 3.7

SOLVING TWO-STEP, REAL-WORLD PROBLEMS WITH FRACTIONS OR DECIMALS – PART 2

LESSON INTRODUCTION

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

LEARNING TARGETS

- I can solve two-step, real-world problems with decimals or fractions.

BENCHMARK COHERENCE

BUILDING ON

MA.5.AR.1 Solve problems involving the four operations with whole numbers and fractions.

WORKING TOWARD

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- What are the similarities and differences between solving one-step and two-step real-world problems?
- What different strategies can be used to solve two-step equations with decimals and fractions?

LESSON NARRATIVE

In this lesson, students continue applying their knowledge of operations on rational numbers to solve two-step problems. After 3.7.1 Warm-Up comparing prices, students complete an Error Analysis task where they analyze and explain the errors, and then do the problem correctly while providing reasoning for their solution. Then, students collaborate by using feedback to improve on their solutions. Students then use the problem-solving process and skills from previous lessons, including converting rational numbers and writing numeric expressions for real-world problems, to solve two-step real-world problems involving decimals and fractions.

COMPONENT SUMMARY

3.7.1 WARM-UP *(~5 MIN.)*

Would You Rather Activity

3.7.2 COLLABORATION *(10-15 MIN.)*

Error analysis, correction, and growth mindset message on two-step, real-world problems

3.7.3 GUIDED PRACTICE *(10-15 MIN.)*

Guided practice solving two-step, real-world problems

3.7.4 YOUR TURN! *(15-20 MIN.)*

Practice solving two-step, real-world problems (optional Gallery Walk opportunity)

3.7.5 WRAP-UP *(~5 MIN.)*

Students solve a two-step problem individually as an assessment of learning targets

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may create inaccurate numeric expressions.
- Students may make computational errors when performing operations on multi-digit decimal and fraction problems.
- Students may incorrectly round their answer for money problems.

THINGS TO CONSIDER

- To foster flexible thinking and problem-solving strategies, ask students to draw a visual model first and then to create a numeric expression from the visual model.
- Students may need to round to the nearest hundredth for money problems so that solutions make sense.
- Encourage students to use problem-solving strategies and not processes that involve using only keywords or shortcuts.

LITERACY AND LANGUAGE ROUTINES

Number Talk

3.7.1 Warm-Up offers a unique opportunity for students to engage in meaningful mathematical discourse in a friendly way – with Skittles. Consider what this entry point (even with real Skittles) could do for activating students' engagement with decimals prior to the lesson.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In 3.7.2 Collaboration, students analyze the mathematical thinking of another student. Consider the clarification language in MA.K12.MTR.4.1 and how that can help frame important mathematical dialogue as students discuss their thinking, sequence and present their findings, and develop methods of comparison and justification when analyzing work.

DIFFERENTIATE INSTRUCTION

For 3.7.4 Your Turn!, questions 1 and 3 use fractions while questions 2 and 4 demonstrate decimal fluency. Consider how to best engage your students in sequencing which questions they work on and in what order. Students could complete one of each before comparing with a peer, or students could engage in choosing which they feel least or most confident in first.

For 3.7.4 Your Turn!, encourage visual learners to draw models prior to creating numerical expressions. For students who are struggling with the computations involving fraction or decimal parts, consider replacing the fractions and decimals with whole numbers, allowing them to focus on understanding the problem-solving process rather than the computation.

3.7.1 WARM-UP: WOULD YOU RATHER?

Component Overview: *Students compare two different deals for Skittles to determine the best purchase. Consider using Think Pair Share literacy routines to give students to work individually then hear others' strategies or ideas in a class discussion.*

TEACHER MOVES

Students should be applying skills of estimation in ranges of fractional or decimal quantities. Consider leveraging some of their responses from this activity later in the lesson when grappling with similar fraction and decimal quantities.

- "I figured there are 8 times as many skittles in the second offer, but the price is less than 8 times the 25 cents." (ratios)
- "I used division to figure out how much 1 skittle cost in each, with the second offer being less cost per skittle." (unit rate)

3.7.1 Warm-Up: Would You Rather?

Would you rather buy 8 Skittles for 25 cents or 62 Skittles for $1.49?

1. **Explain your choice.**

   Answers will vary.

2. **Would your answer change if you could choose what flavor the 8 Skittles were but you could not choose the flavors from the bag?**

   Answers will vary.

3.7.2 Collaboration: Error Analysis

1. Stefanie solved the following word problem. Her work is shown.

   a. Evaluate her work and determine where she made mistakes.

Problem	Stefanie's Work
Janelle has $2\frac{1}{4}$ pans of brownies. Each pan made 8 brownies. She gave $\frac{2}{3}$ of the brownies to her friends. How many brownies did Janelle give away?	Step 1: $2\frac{1}{4} \times 8 = \frac{9}{4} \times 8 = \frac{18}{4}$ Step 2: $\frac{18}{4} \div \frac{2}{3} = \frac{18}{4} \times \frac{3}{2} = 6\frac{3}{4}$ Answer: She gave away $6\frac{3}{4}$ brownies.

   b. Help Stefanie understand and learn from her mistakes. Write a brief note to Stefanie explaining what her mistakes were and how she could learn from those mistakes.
   Answers may vary.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Possible answer:

When finding a fraction of an amount, use multiplication. When multiplying a fraction by a whole number, change the whole number to a fraction with denominator of 1.

Solution: $\left(2\frac{1}{4} \times 8\right) \times \frac{2}{3} = 12$

   c. Read your partner's note to Stefanie and offer suggestions for how to improve their note.

3.7.2 COLLABORATION: ERROR ANALYSIS

Component Overview: *Students complete an Error Analysis task where they analyze and explain the errors, then provide a brief note explaining the mistake and how to learn from it. Group students in partnerships as they evaluate the work, then complete the problem correctly and provide reasons for their solution. Consider combining partnerships for the feedback in question 1c so students share their thinking with others.*

TEACHER MOVES

MA.K12.MTR.4.1: Error Analysis

Stefanie makes a computational error in Step 1 when determining how many total brownies are available to give away. Encourage students not only to identify the error, but also to highlight the correct use of multiplication to determine the number of brownies, as well as the opportunity to assess that $\frac{18}{4}$ brownies was not reasonable given the context.

ENRICHMENT

Consider the following probing questions as students collaborate on identifying the error:

- Why is $2\frac{1}{2}$ multiplied by 8? Does the answer $\frac{18}{4}$ make sense? Why or why not?
- Why is that product then divided by $\frac{2}{3}$? Why not addition or multiplication?
- Does the answer seem reasonable?

3.7.3 GUIDED PRACTICE: TWO-STEP RATIONAL PROBLEMS

Component Overview: *Students work individually or in partnerships to solve additional two-step problems involving rational numbers. Consider creating a teacher-led small group for students who may need additional scaffolding or support.*

TEACHER MOVES

Think-Pair-Share

Pull an optional teacher-led small group and review the usefulness of visual models or representations when operating with fractions or decimals. How could a visual model be used in question 1 that might better help students connect the difference between $\frac{1}{2}$ and $\frac{5}{12}$?

QUESTIONING

Ask students who finish early to extend question 2 by finding the cost to travel a certain number of miles or kilometers using the gas prices given for at least two different mileage efficiencies.

3.7.3 Guided Practice: Two-Step Rational Problems

1. While birdwatching, David saw a cardinal $\frac{1}{2}$ of a yard to the left from where he was standing. A bluebird was $\frac{5}{12}$ of a yard to the right of the cardinal. Crystal is standing $1\frac{1}{3}$ yards to the right of David. How far was the bluebird from where Crystal was standing?
   $1\frac{5}{12}$ of a yard

2. One gallon of gasoline in Buffalo, New York costs $2.29. In Toronto, Canada, one liter of gas is $0.91. There are 3.8 liters in one gallon. How much greater is the cost of gas in either city?
   Gas is $1.17 greater in Toronto.

3.7.4 Your Turn!

1. Martha fills her ice cream bowl with $\frac{3}{4}$ cup of ice cream, $\frac{1}{8}$ cup of candy toppings, and the rest with fruit toppings. If the bowl contains a total of $1\frac{1}{2}$ cups, how much of the bowl is made up of fruit toppings?
   $\frac{5}{8}$ cups of fruit topping

2. Jasmine has a ribbon that is 12.6 meters long. Sharon has a ribbon that is 3.8 meters shorter. What is the total length of the two ribbons?
   21.4 meters

3. Before soccer practice, Hernando puts $4\frac{1}{8}$ cups of sports drink in a water bottle with $1\frac{2}{3}$ cups of ice. After a hot afternoon of practice, he drinks $2\frac{1}{5}$ cups of the sports drink and melted ice. How many cups of this mixture are left in the water bottle?
   $3\frac{71}{120}$ cups

4. A bag of chocolates weighs 1.384 pounds. If Luis gives each of his 6 friends 0.23 pounds of chocolate, how many pounds of chocolate does Luis have left?
   0.004 pounds

3.7.4 YOUR TURN!

Component Overview: *Consider using a Gallery Walk for this activity. Provide small groups of 2 to 4 with chart paper and markers. Have students solve all four problems on their chart paper. After about 15 minutes of small group work time, have students display their chart paper on a visible wall/desk, then have the class walk around to view other groups' posters for about 5 minutes. Ask students to visit at least three posters and confirm the work on the poster with their own answer and compare strategies used.*

DIFFERENTIATE INSTRUCTION

Encourage students to draw visual models prior to attempting to solve the problem. This may help students make connections from a pictorial representation to the numeric expression and operation needed to solve.

TEACHER MOVES

To encourage mathematical reasoning, ask questions like "Do your answers make sense? What methods can you use to check your answer?" (MA.K12.MTR.7.1)

ENRICHMENT

As students work on solving these problems, circulate the room and listen for how students are making sense of these problems. To develop strategies for understanding problems prior to solving, ask questions like:

- How do you know what you are being asked to find?
- Which step/operation do you need to use first? Which operation/step do you use second? How do you know?
- What strategy can you use? Does drawing a model make sense for this problem?
- Is your answer reasonable? How do you know?

3.7.5 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *This Wrap-Up assesses students' ability to solve two-step problems involving rational numbers. Students individually solve the two-step problem as a formative assessment on solving two-step problems from this and the previous lesson. Students should not use a calculator to complete the Wrap-Up.*

DIFFERENTIATE INSTRUCTION

Encourage visual learners to draw visual models prior to creating numerical expressions, particularly referencing the ribbon activity from the previous lesson as an anchor.

3.7.5 Wrap-Up: Ticket Out the Door

1. Priscilla and Dayton sold candy to raise money for their swimming team. Priscilla sold $1\frac{5}{8}$ times as much as Dayton. If Dayton sold $\frac{1}{5}$ of a box of candy, how many total boxes of candy did they sell?

   $\frac{21}{40}$ *boxes*